Reading group notes: mixture-of-experts

Reading group of the Ashbury machine learning lab

1 Background

Earlier mixture-of-experts methods improved the router but left the top-k routing unchanged. Recent work has made the shared expert a central concern, because it decides how far mixture-of-experts models can scale. We argue that the load balance deserves as much attention as the token dropping. Most prior methods treat the gating network as fixed and tune the token dropping around it.

Combining mixture-of-experts with methods that target the capacity factor is a promising direction. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. We expect the gating network to interact with the expert parallelism at larger scales, which we leave to future work. The load balance also makes mixture-of-experts easier to inspect, since its behaviour can be read off per token.

Practitioners report that the auxiliary loss is the first component to fail on long inputs. Despite steady progress, the interaction between the token dropping and the capacity factor is still poorly understood. This work studies mixture-of-experts, a method that rethinks the gating network. Earlier mixture-of-experts methods improved the capacity factor but left the shared expert unchanged.

2 Methods we read

The shared expert is initialised so that mixture-of-experts starts training close to a dense model. Mixture-of-experts computes the auxiliary loss once per layer and reuses it for every head. Unlike a dense model, mixture-of-experts updates the expert only when the gating network changes. A coefficient controls how strongly the top-k routing depends on the expert; we set it to 0.4 throughout.

Mixture-of-experts computes the expert once per layer and reuses it for every head. During inference mixture-of-experts caches the shared expert, so each new token costs a constant amount of work. We implement the router as a fused kernel, which avoids writing intermediate results to memory. Unlike switch routing, mixture-of-experts updates the shared expert only when the gating network changes.

The router in mixture-of-experts is parameterised by a small linear layer with 64 hidden units. Each block of mixture-of-experts applies the load balance after a short convolution of width 2. The capacity factor is initialised so that mixture-of-experts starts training close to switch routing. A coefficient controls how strongly the top-k routing depends on the capacity factor; we set it to 0.9 throughout.

3 Results we discussed

Variance across seeds stays below 0.4 points for every configuration we report. Mixture-of-experts scores 48.7% on WebMix perplexity, compared with 51.1% for a top-1 router. The gains of mixture-of-experts grow with the input length, as the top-k routing becomes the bottleneck. With the expert parallelism frozen, mixture-of-experts still recovers most of its improvement on a reasoning suite.

Mixture-of-experts scores 63.7% on a translation suite, compared with 42.0% for a dense model. Variance across seeds stays below 0.9 points for every configuration we report. With the expert parallelism frozen, mixture-of-experts still recovers most of its improvement on a reasoning suite. We train every model for 70B tokens with the same data order and report the mean of three seeds.

On WebMix perplexity, mixture-of-experts improves over switch routing by 2.1 points. We train every model for 40B tokens with the same data order and report the mean of three seeds. Throughput is measured with a batch size of 256 on a single accelerator. Ablating the expert lowers the WebMix perplexity score by 3.4 points, the largest drop among all components.

4 Open questions

A limitation of mixture-of-experts is that the load balance must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The load balance also makes mixture-of-experts easier to inspect, since its behaviour can be read off per token. The results suggest that the token dropping is the main lever for quality at a fixed budget.

We expect the gating network to interact with the shared expert at larger scales, which we leave to future work. A limitation of mixture-of-experts is that the auxiliary loss must be tuned separately for each model size. Our analysis covers models up to 1.3B parameters, and larger models may behave differently. The results suggest that the auxiliary loss is the main lever for quality at a fixed budget.